Solving Chess via Higher-Dimensional Lattice Projections, Algebraic Ray Interdiction Operators, and Topological Constraint Propagation

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Abstract

Traditional computer chess engines rely on brute-force alpha-beta pruning over massive state-space trees, treating the board as a flat Euclidean grid $\mathbb{Z}^2$. In this white paper, we formalize an alternative paradigm developed at Glover Labs: modeling chess as a higher-dimensional hyper-cubic lattice projection, where piece movements are orthogonal linear transformations rather than non-linear discontinuities. We introduce the algebraic **Ray Interdiction Operator (I)** to replace iterative loop-based ray casting with eigenvalue collapse, and we formalize **Topological Constraint Propagation** to prune combinatorially explosive sub-trees via formal algebraic invariants.

1 Introduction and Motivation

Classical chess engines (e.g., Stockfish) evaluate positions through depth-first minimax search coupled with heuristic evaluation functions. While computationally formidable, this approach suffers from a combinatorial explosion of states ($\approx 10^{120}$ possible game paths).

Human insight and mathematical elegance suggest a different path: treating the chess board not as a static board, but as a dynamic algebraic manifold. By lifting the state space into a higher-dimensional lattice and defining piece interactions as vector intersections, we can collapse complex tactical configurations into linear operations.

2 The Higher-Dimensional Lattice Projection Space ($\mathbb{Z}^6 \to \mathbb{Z}^2$)

In standard 2D Euclidean space ($\mathbb{Z}^2$), certain piece mechanics—such as the Knight's L-jump or diagonal sliding vectors—appear geometrically disjoint or non-linear. To linearize these movements, we embed the 8×8 board into a 6-Dimensional Hyper-Lattice $\mathbb{Z}^6$, where each fundamental movement vector corresponds to an orthogonal coordinate axis:

$$\begin{aligned} &\text{Axis } 1, 2: && \text{Orthogonal space } (X, Y) \text{ for Rooks and Queens} && (1)\\ &\text{Axis } 3, 4: && \text{Diagonal space } (D_1, D_2) \text{ for Bishops and Queens} && (2)\\ &\text{Axis } 5, 6: && \text{L-jump space } (L_1, L_2) \text{ for Knights} && (3) \end{aligned}$$

Let the mapping function Φ project higher-dimensional integer coordinates down to the physical board:

$$\Phi : \mathbb{Z}^6 \longrightarrow \mathbb{Z}^2$$

By applying a Cut-and-Project transformation matrix **P** (incorporating irrational slope factors inspired by de Bruijn's aperiodic tilings), complex multi-piece tactical forks and pins become simple geometric intersections of hyperplanes:

$$\mathbf{X}_{\text{board}} = \mathbf{P} \cdot \mathbf{Z}_{\text{6D}}$$

3 Algebraic Ray Interdiction Operators (I)

Rather than looping through square offsets to compute ray attacks, we treat ray interdiction as a matrix projection and occlusion operator.

Let a ray R originating at coordinate $\mathbf{p}_0$ sweep across an ordered set of squares:

$$S_{\text{ray}} = \{\mathbf{p}_0 + k\mathbf{v} \mid k \in \mathbb{N}, 1 \leq k \leq K\}$$

We define the diagonal Interdiction Matrix $\mathbf{M}_{\text{int}}$ along the ray vector:

$$\mathbf{M}_{\text{int}}(k) = \begin{cases} 1 & \text{if square } \mathbf{p}_k \text{ is empty (transparent)} \\ 0 & \text{if square } \mathbf{p}_k \text{ contains any piece (occluded)} \end{cases} \tag{4}$$

The effective reach matrix $\mathbf{R}_{\text{eff}}$ is computed via cumulative matrix multiplication:

$$\mathbf{R}_{\text{eff}} = \prod_{k=1}^{m} \mathbf{M}_{\text{int}}(k) \cdot \mathbf{V}_{\text{base}}$$

When a piece occupies coordinate k, $\mathbf{M}_{\text{int}}(k) = 0$, causing all downstream eigenvalues (squares further along the ray) to instantly collapse to zero without iterative search loops.

4 Topological Constraint Propagation and Invariants

As demonstrated by our topological analysis of piece mobility, every exit path from a square is symmetrically a re-entrance path back to it. Let B be a linear-ray piece at square S. Any egress move $M : S \to S'$ along vector $\mathbf{w}$ maintains an inverse re-entrance vector $\mathbf{u} = -\mathbf{w}$.

By encoding these vector fields as formal axioms, Glover Labs' framework propagates constraints across the board: 1. **Empire Bounds:** Local piece placements instantly restrict legal move sub-graphs globally across non-local vector rays. 2. **Pruning via Invariants:** Invalid tactical states are discarded at the algebraic level before entering the search tree.

5 Conclusion

By combining higher-dimensional lattice projections with algebraic ray interdiction and topological constraint propagation, we transition chess from an exhaustive search problem to a structured geometric-algebraic system. Future work at Glover Labs will focus on implementing these sparse matrix operators in custom tensor hardware to test analytical solvability bounds.

References

[1] Penrose, R. (1989). *The Emperor's New Mind: Concerning Computers, Minds, and the Laws of Physics*. Oxford University Press.

[2] de Bruijn, N. G. (1981). Algebraic theory of Penrose's non-periodic tilings of the plane. *Indagationes Mathematicae*, 84(1), 39-66.